

FUNDAMENTALS OF PROJECT COST CONTROL

Step No.	Cost-Control Method	What to Do	Why It Matters	Target / Control Measure
1.	Establish the Cost Baseline	Approve the time-phased project budget before execution begins.	Provides the official reference for measuring cost performance.	100% approved before work starts.
2.	Develop a Cost Breakdown Structure	Divide the total budget into work packages, activities, resources, materials, and accounts.	Makes cost ownership and tracking easier.	Every cost linked to a CBS code.
3.	Link Costs to the Work Breakdown Structure	Assign each cost item to the corresponding WBS activity.	Connects scope, schedule, and budget control.	100% cost-to-WBS alignment.
4.	Define Cost Control Accounts	Create control accounts for departments, contractors, phases, and major deliverables.	Improves accountability and performance measurement.	One responsible owner per account.
5.	Track Budget Versus Actual Cost	Compare planned expenditure with actual expenditure at regular intervals.	Identifies overspending before it becomes critical.	Cost variance within $\pm 5\%$.
6.	Calculate Cost Variance	Use the formula $CV = EV - AC$.	Shows whether completed work is under or over budget.	$CV \geq 0$.
7.	Monitor Cost Performance Index	Use the formula $CPI = EV \div AC$.	Measures the cost efficiency of completed work.	$CPI \geq 1.00$.
8.	Update the Cost Forecast	Calculate the expected total cost at project completion.	Provides early warning of a potential budget overrun.	$EAC \leq$ approved budget.
9.	Calculate Estimate to Complete	Use the formula $ETC = EAC - AC$.	Shows the remaining funding required to finish the project.	Realistic and regularly updated.
10.	Monitor Committed Costs	Track purchase orders, contracts, approved invoices, and pending obligations.	Prevents hidden financial commitments from being overlooked.	Actual plus committed cost within budget.
11.	Control Scope Changes	Evaluate the cost impact of every requested change before approval.	Prevents uncontrolled scope growth and budget leakage.	No work without approved change authorization.
12.	Manage Contingency Reserves	Release contingency funds only for approved and documented risks.	Protects reserves from unnecessary consumption.	Reserve usage aligned with risk exposure.
13.	Review Procurement Costs	Compare supplier quotations, contract rates, and market prices.	Reduces purchasing costs and improves commercial value.	Procurement variance within approved limits.
14.	Control Material Waste	Measure ordered, consumed, damaged, returned, and unused materials.	Reduces waste, theft, rework, and storage costs.	Waste percentage below project threshold.
15.	Monitor Labor Productivity	Compare actual labor hours and output with the approved productivity baseline.	Identifies inefficient working practices and overtime exposure.	Productivity equal to or above baseline.
16.	Improve Equipment Utilization	Track productive hours, idle time, maintenance, and rental duration.	Prevents unnecessary equipment and rental expenses.	Utilization $\geq 85\%$.
17.	Reduce Rework Costs	Record the cost of defects, rejected work, repairs, and repeated activities.	Protects both the project budget and schedule.	Rework cost $\leq 5\%$ of contract value.
18.	Validate Invoices and Progress Claims	Confirm quantities, rates, approvals, and completed work before payment.	Prevents duplicate payments and inaccurate billing.	100% invoice verification.
19.	Conduct Monthly Cost Reviews	Review variances, commitments, cash flow, risks, forecasts, and corrective actions.	Keeps management informed and supports timely decisions.	One formal cost review every month.
20.	Implement Corrective Action	Assign owners, deadlines, and recovery measures when costs deviate from the baseline.	Turns cost information into measurable financial improvement.	Corrective action started within seven days.



Essential Project Cost-Control Formulas

1 Cost Variance:

$$CV = EV - AC$$

4 Estimate to Complete:

$$ETC = EAC - AC$$

2

Cost Performance Index:

$$CPI = EV \div AC$$

5 Variance at Completion:

$$VAC = BAC - EAC$$

3

Estimate at Completion:

$$EAC = BAC \div CPI$$

6 To-Complete Performance Index:

$$TCPI = \text{Remaining Work} \div \text{Remaining Budget}$$



Review Rule:

Review cost performance weekly at the operational level and monthly at the management level.



Update Rule:

Update actual costs, commitments, forecasts, risks, and approved changes before issuing the cost report.



Key Success Tips:

- Assign a cost owner to every control account.
- Integrate scope, schedule, risk, procurement, and cost data.
- Use dashboards to identify negative trends early.
- Investigate the root cause of every major variance.
- Never approve additional work without confirming its budget impact.
- Use corrective and preventive actions before contingency funds are exhausted.
- Maintain one reliable source of financial information.



Control the cost before the cost controls the project.

rPlan

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Dwg Purpose Code :-

Status of Approval :-

Dimensions as Mentioned

Engg Code :-

Backup Info Code :-

Angle of Projection :-

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Dwg electronic code

Drawn By:

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Description:

PROJECT COST CONTROL

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SCALE:-

SHEET